The Blessing of Dimensionality: How Near-Orthogonality in High-Dimensional Spaces Explains Temporal Portability

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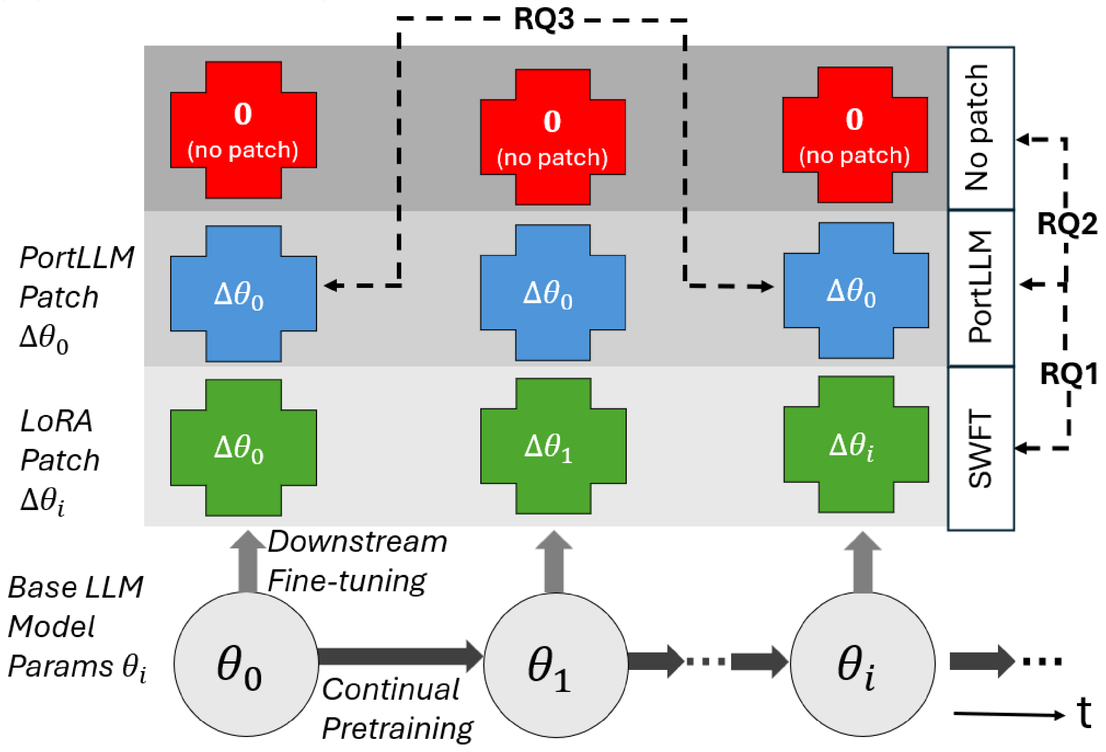
Motivation and Merits

LLMs may be repeatedly updated via continual pretraining. Is repeated fine-tuning needed? PortLLM eliminates this need

We find: (1) PortLLM is effective for at least 10 updates
(2) Near-orthogonality explains this

Research Questions

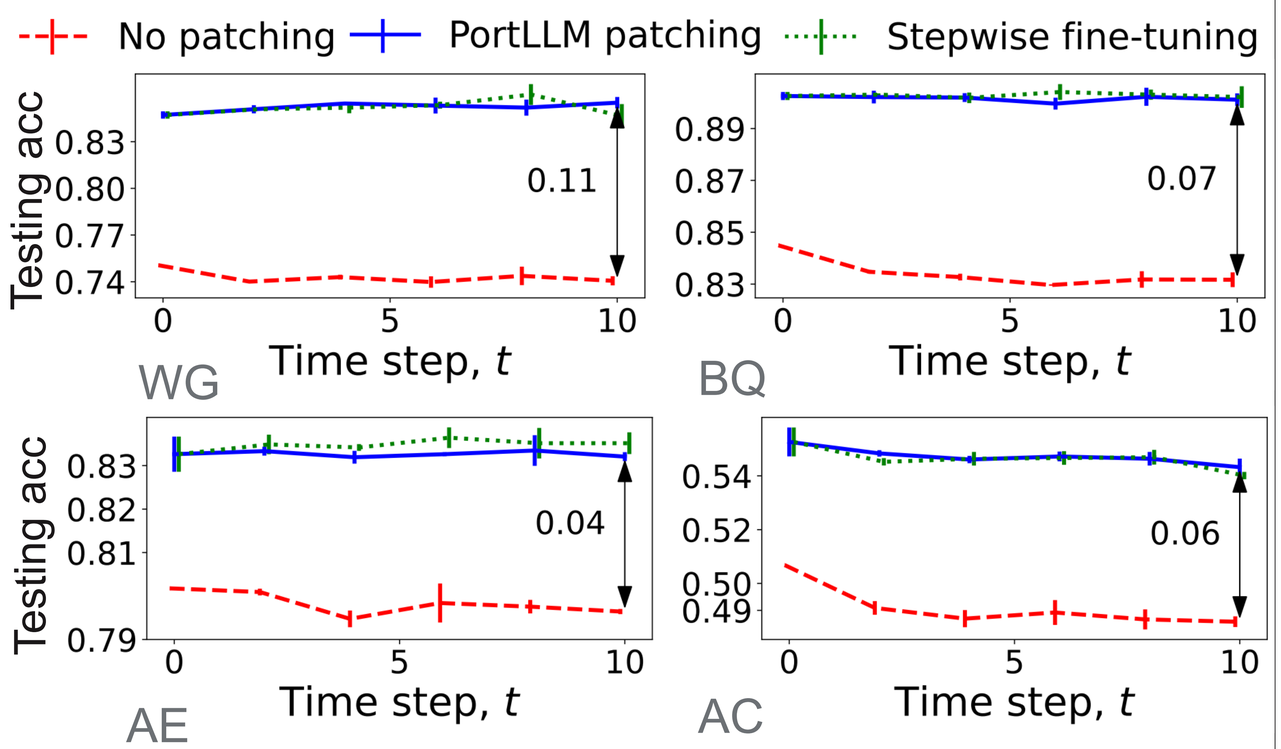
Model update at each time step via continual pretraining
3 Patching strategies:
(1) **Stepwise fine-tuning** (SWFT): Fine-tune new LoRA patch [1] at each time step
(2) **PortLLM** [2]: Retain initial LoRA patch from $t = 0$
(3) **No patching**: Use no LoRA patch



RQ1: (1) – (2), **RQ2**: (2) – (3)
RQ3: Temporal variation in (2)

[1] Hu, E. J., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., & Chen, W. LoRA: Low-Rank Adaptation of Large Language Models. In *ICLR*, 2021.
[2] R. M. S. Khan, P. Li, S. Yun, Z. Wang, S. Nirjon, C.-W. Wong, and T. Chen. PortLLM: Personalizing evolving LLMs with training-free and portable model patches. In *ICLR*, 2025.

Pilot Results



RQ1: PortLLM and SWFT are comparable
RQ2: PortLLM outperforms no patching
RQ3: PortLLM is approximately constant across time steps
Using Mistral-7B-v0.1 continually pretrained on Fineweb
WinoGrande (WG), BoolQ (BQ), ARC-Easy (AE), ARC-Challenge (AC)

Proposed Math Tool: DGD

Definition: $F(\gamma\Delta x) = \langle \nabla f(x_0 + \gamma\Delta x), \gamma\Delta x \rangle$

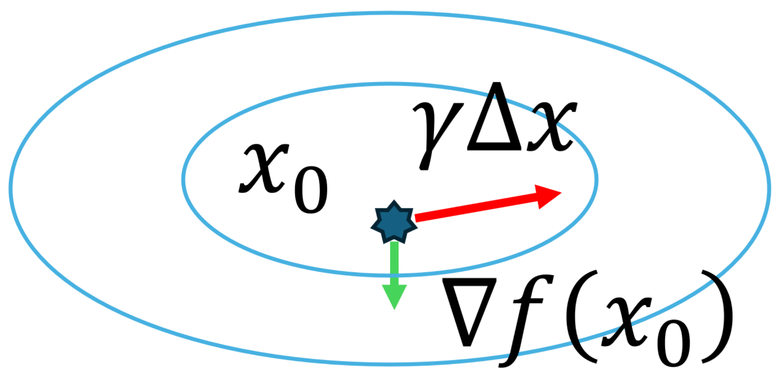
Displaced gradient (DG) Displacement (D)

Bound: $F(\gamma\Delta x) \leq \gamma\langle \nabla f(x_0), \Delta x \rangle + \gamma^2 \sigma_{\max}(x_0 + \beta'\gamma\Delta x)$

Alignment Sharpness

If gradient $\approx\perp$ displacement, then small alignment

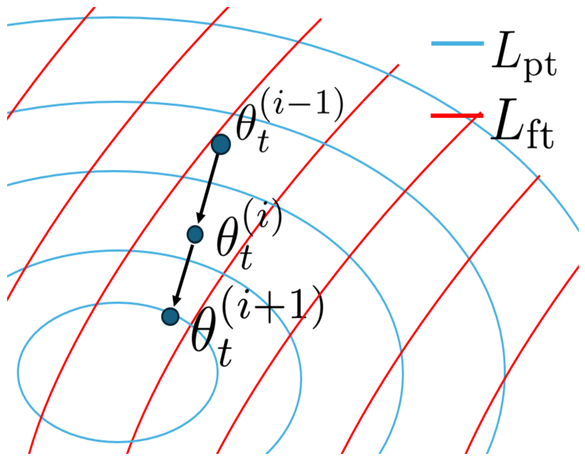
$\approx\perp$ expected in high-dimensional spaces



Analysis from Training Iterates

Proof Idea:
Gradient descent per pretraining loss L_{pt}

Track performance per fine-tuning loss L_{ft}

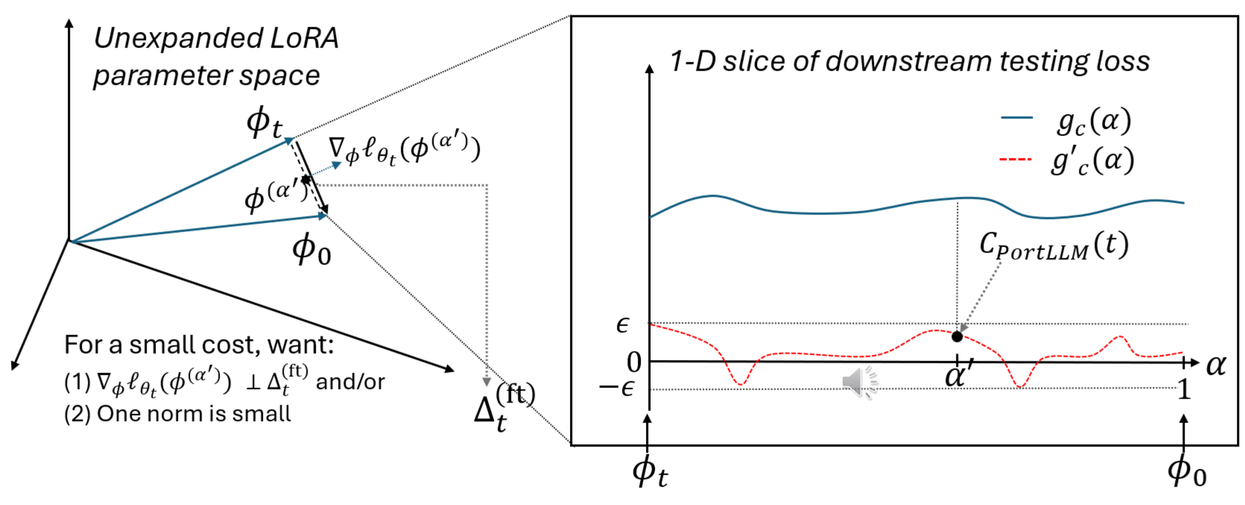


Bound on variation (RQ3):

$$|L_{\text{ft}}(\theta_t + \Delta\theta_0) - L_{\text{ft}}(\theta_0 + \Delta\theta_0)| \leq It(\eta\epsilon_{orth}\ell_{\text{pt}}\ell_{\text{ft}} + \epsilon_\eta)$$

ϵ_{orth} is a quasi-orthogonality constant
I is # of continual pretraining opt. steps, η is learning rate, ℓ_{pt}, ℓ_{ft} and ϵ_η are bounds
Main Assumption: Pretraining and fine-tuning gradients are nearly orthogonal so that their cosine similarity is upper bounded by ϵ_{orth}

Analysis from 1-D Slice



Loss on 1-D slice: $g_c(\alpha) = \ell_{\theta_t}(\alpha\phi_0 + (1-\alpha)\phi_t)$
Cost of PortLLM: $C_{\text{PortLLM}} = \ell_{\theta_t}(\phi_0) - \ell_{\theta_t}(\phi_t)$
$C_{\text{PortLLM}} = g_c(1) - g_c(0) = g'_c(\alpha') = \langle \nabla \ell_{\theta_t}(\phi^{(\alpha)}), \Delta_t^{\text{ft}} \rangle$